

**Choose the right layer:** If already on Btrfs for backup disks, use *duperemove* weekly or *bees* continuously on that subvolume. Start with a test directory to measure space savings and runtime.

If you want filesystem-agnostic dedup on a SAN or local RAID, evaluate *dm-vdo* on a new test LUN, layer XFS or ext4 on top, then replay a typical backup workload and compare logical vs physical use.

If you are running ZFS for VM pools, enable dedup only on datasets that contain highly similar data. Monitor DDT size, RAM use, and read latency over a few weeks before scaling out.

**Combine dedup with smart backup design:** Hybrid designs from systems like RevDedup show that you don't have to choose between performance and savings.

A simple adaptation for Linux is:

- Take regular snapshots on a CoW filesystem (Btrfs/ZFS).
- Deduplicate older snapshots more aggressively, leaving the newest generation relatively unfragmented.
- When you prune snapshots, prefer to delete heavily fragmented, older generations.

This pushes fragmentation cost towards data that is rarely read and keeps recent backups fast.

**Watch the numbers that actually matter:** Track:

- Logical vs physical usage (your effective dedup + compression ratio)
- Average and tail latencies for backup and restore jobs
- Dedup index size and memory footprint (DDT size for ZFS, index size for VDO, database size for duperemove)

If your dedup ratio is below ~1.2:1 and you are paying a heavy performance or complexity tax, it may not be worth it for that workload.

## Where Linux dedup is heading

Recent research and kernel work hints at where Linux storage is going next:

- Faster content-defined chunking (e.g., RapidCDC and newer CDC variants) that keep high dedup ratios while slashing CPU cost.
- Inline dedup optimised for non-volatile memory and SSDs, where random I/O penalties are smaller and metadata can live in persistent memory.
- Container-aware dedup engines that understand Docker or OCI layer semantics and avoid redundant copies across thousands of images without killing pull performance.

For practitioners, this means dedup will increasingly feel less like an expensive experiment and more like a predictable building block, especially in backup, VM, and container-image platforms. **END** 🐧

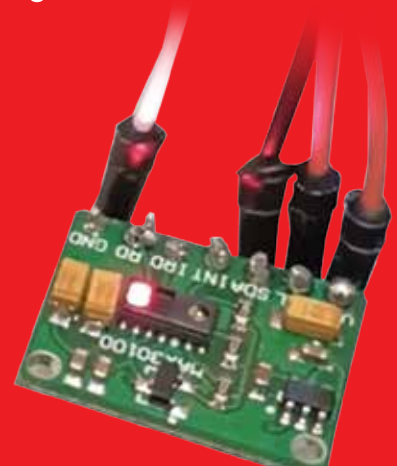
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